

# Complex Numbers

## Cartesian Form

$$z = x + iy \quad \bar{z} = x - iy$$

$$\operatorname{Re} z = x = \frac{1}{2}(z + \bar{z}) \quad \operatorname{Im} z = y = \frac{1}{2i}(z - \bar{z})$$

$$\overline{(z_1 + z_2)} = \bar{z}_1 + \bar{z}_2 \quad \overline{(z_1 - z_2)} = \bar{z}_1 - \bar{z}_2$$

$$\overline{(z_1 z_2)} = \bar{z}_1 \bar{z}_2 \quad \overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$$

## Polar Form

$$x = r \cos \theta \quad y = r \sin \theta \quad z = r(\cos \theta + i \sin \theta)$$

$$|z| = r = \sqrt{x^2 + y^2} = \sqrt{z\bar{z}}$$

$$\arg z = \theta \quad -\pi < \operatorname{Arg} z \leq \pi$$

$$\tan \theta = \frac{y}{x}$$

$$z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$$

$$|z_1 z_2| = |z_1| |z_2| = r_1 r_2 \quad \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|} = \frac{r_1}{r_2}$$

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2$$

$$\arg \frac{z_1}{z_2} = \arg z_1 - \arg z_2 \quad \text{up to multiples of } 2\pi$$

$$z^n = r^n (\cos n\theta + i \sin n\theta)$$

$$\sqrt[n]{z} = \sqrt[n]{r} \left( \cos \frac{\theta + 2k\pi}{n} + i \sin \frac{\theta + 2k\pi}{n} \right)$$

## Graphing

Circle with radius  $R$  and center  $z_0$ :

$$|z - z_0| = R$$

Disc with radius  $R$  and center  $z_0$ :

$$|z - z_0| < R$$

Annulus with outer radius  $R$ , inner radius  $r$ , and center  $z_0$ :

$$r < |z - z_0| < R$$

Wedge between  $-\theta$  and  $\theta$ :

$$|\arg z| < \theta$$

Half plane bounded at  $z = a$ :

$$\operatorname{Re} z \geq -1$$

# Analytic Functions

Functions that are differentiable in *some* domain. Functions are considered **entire** if differentiable *everywhere*.

## Test via Cauchy-Riemann Equations

$$f(x) = u(x, y) + iv(x, y) = u(r, \theta) + iv(r, \theta)$$

$$u_x = v_y \quad \text{and} \quad u_y = -v_x$$

$$u_r = \frac{1}{r} v_\theta \quad \text{and} \quad v_r = -\frac{1}{r} u_\theta$$

## Harmonic Functions

$$\Delta u = 0$$

Use Cauchy-Riemann Equations to calculate a harmonic conjugate function  $v$  and find  $f(z)$ .

## Derivatives

$$f'(z) = u_x + iv_x = v_y - iu_y$$

## Exponential

$$e^z = e^x e^{iy} = e^x (\cos y + i \sin y)$$

$$|e^z| = e^x$$

## Solving for z

$$e^z = e^x (\cos y + i \sin y) = a + ib$$

$$|e^z| = e^x = \sqrt{a^2 + b^2} \implies x = \ln \sqrt{a^2 + b^2}$$

$$y = \arg(a + ib) = \tan^{-1} \left( \frac{b}{a} \right) + 2n\pi, \quad (n \in \mathbb{Z})$$

## Trigonometry

$$\cos z = \frac{1}{2}(e^{iz} + e^{-iz}) \quad \sin z = \frac{1}{2i}(e^{iz} - e^{-iz})$$

$$\cosh z = \frac{1}{2}(e^z + e^{-z}) \quad \sinh z = \frac{1}{2i}(e^z - e^{-z})$$

$$\cos iz = \cosh z \quad \sin iz = -i \sinh z$$

$$\cosh iz = \cos z \quad \sinh iz = -i \sin z$$

$$\cos z = \cos x \cosh y - i \sin x \sinh y$$

$$\sin z = \sin x \cosh y + i \cos x \sinh y$$

$$|\cos z|^2 = \cos^2 x + \sinh^2 y$$

$$|\sin z|^2 = \sin^2 x + \sinh^2 y$$

## Identities

$$\sin^2 \theta = \frac{1}{2} - \frac{1}{2} \cos 2\theta$$

$$\cos^2 \theta = \frac{1}{2} + \frac{1}{2} \cos 2\theta$$

$$\sin \theta \cos \theta = \frac{1}{2} \sin 2\theta$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

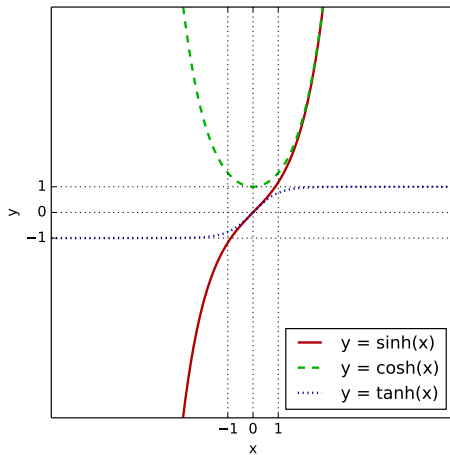
$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$$

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$$

## Values



|               | 0 | $\pi/6$      | $\pi/4$      | $\pi/3$      | $\pi/2$ |
|---------------|---|--------------|--------------|--------------|---------|
| $\sin \theta$ | 0 | $1/2$        | $\sqrt{2}/2$ | $\sqrt{3}/2$ | 1       |
| $\cos \theta$ | 1 | $\sqrt{3}/2$ | $\sqrt{2}/2$ | $1/2$        | 0       |
| $\tan \theta$ | 0 | $1/\sqrt{3}$ | 1            | $\sqrt{3}$   | —       |
| $\sinh$       | 0 |              |              |              |         |
| $\cosh$       | 1 |              |              |              |         |
| $\tanh$       | 0 |              |              |              |         |

## Logarithm

$$\operatorname{Ln} z = \ln |z| + i \operatorname{Arg} z$$

$$\ln z = \ln r + i\theta = \operatorname{Ln} z + 2n\pi i, \quad (n \in \mathbb{Z})$$

$$z^c = e^{c \ln z} \quad a^z = e^{z \ln a}$$

## Line Integrals

### Analytic Functions

$$\int_C f(z) dz = \int_{z_0}^{z_1} f(z) dz = F(z_1) - F(z_0)$$

## General

1. Represent the path  $C$  in the form  $z(t)(t \in [a, b])$
2. Calculate the derivative  $\dot{z}(t)$
3. Substitute  $z(t)$  into  $f(z)$
4. Integrate

$$\int_C f(z) dz = \int_a^b f(z(t)) \dot{z}(t) dt$$

## Loop Integrals

$$\oint_C (z - z_0)^m dz = \begin{cases} 2\pi i & (m = -1) \\ 0 & (m \neq -1, m \in \mathbb{Z}) \end{cases}$$

## Analytic Functions $f(z)$

### Cauchy's Integral Theorem

$$\oint_C f(z) dz = 0$$

### Cauchy's Integral Formula

$$\oint_C \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0)$$

$$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz$$

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz \quad (n \in \mathbb{Z}^+)$$

## Series Convergence

### Tests

#### Comparison

If  $|z_i| \leq b_i \forall i$  and the  $b$ -series converges, the  $z$  series converges absolutely.

#### Geometric Series

$$\sum_{n=0}^{\infty} q^n = \begin{cases} \frac{1}{1-q} & |q| < 1 \\ \text{diverges} & |q| \geq 1 \end{cases}$$

#### Ratio Test & Root Test

$$\lim_{n \rightarrow \infty} \left| \frac{z_{n+1}}{z_n} \right| = L \quad \lim_{n \rightarrow \infty} \sqrt[n]{|z_n|} = L$$

- $L < 1 \implies$  converges absolutely
- $L = 1 \implies$  inconclusive
- $L > 1 \implies$  diverges

#### Radius of Convergence

$$|z - z_0| < R = \frac{1}{L}$$